FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – III

JEE (Advanced)-2021

PAPER –1

TEST DATE: 04-07-2021

Time Allotted: 3 Hours **Maximum Marks: 198**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

Section-A (01 – 06, 19 – 24, 37– 42) *contains* ***18 multiple*** *choice questions which have* ***ONLY ONE CORRECT ANSWER****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Section-A (07 – 12, 25 – 30, 43 – 48) *this section contains* ***18 multiple*** *choice questions.*
Each question has FOUR options. ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct answer(s).*
For each question, choose the option(s) corresponding to (all) the correct answer(s)
Answer to each question will be evaluated according to the following marking scheme:

Full Marks **: +4** If only (all) the correct option(s) is (are) chosen:
Partial Marks **: +3** If all the four options are correct but ONLY three options are chosen;
Partial Marks **: +2** If three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks **: +1** If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks **: 0** If none of the options is chosen (i. e. the question is unanswered);
Negative Marks **: –2** In all other cases.

Section-C (13 – 18, 31– 36, 49 – 54) *contains* ***18 Numerical*** *answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***0 marks*** *for wrong answer.*

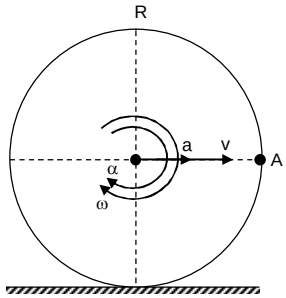
Physics

PART – I

SECTION – A
(One Options Correct Type)

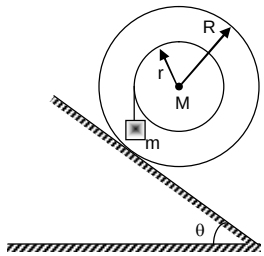
This section contains **06 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A uniform circular disc of radius 'R' is rolling without slipping on a rough horizontal surface with a constant acceleration 'a'. Then the radius of curvature of trajectory of point 'A' of the disc relative to the ground at the given instant as shown in the figure is



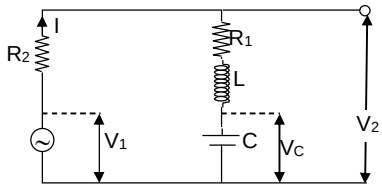
(A) 4R
(B) $2\sqrt{2}R$
(C) 2R
(D) $\sqrt{2R}$

2. A bobbin of mass M = 3kg consists of a central cylinder of radius r = 5 cm and two end plates of radius R = 6 cm. It is placed on a slotted incline on which it will roll but not slip, and a mass m = 4.5 kg is suspended from a cord wound around the bobbin, as shown in figure. It is observed that the system is in static equilibrium. What is the angle of tilt θ of the incline?

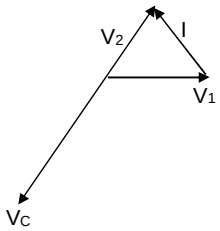


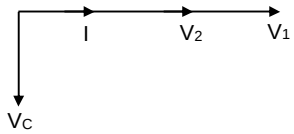
(A) $\theta = 30^0$
(B) $\theta = 45^0$
(C) $\theta = 60^0$
(D) $\theta = 75^0$

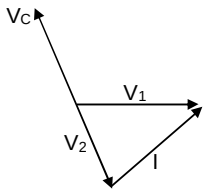
3. The circuit shown in figure is energized by a sinusoidal voltage source V_1 at a frequency which causes resonance with a current of I.

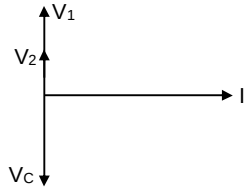


The phasor diagram which is applicable is

(A)


(B)


(C)


(D)


4. A ray of light moving along vector $\left(5\sqrt{2}\hat{i}-5\hat{j}-5\hat{k}\right)$ undergoes refraction at an interface of two media, which is y-z plane. The refractive index for $x \le 0$ is $\sqrt{3}$ while for $x \ge 0$ it is $\sqrt{2}$. Then the unit vector along the refracted ray is

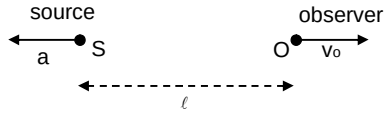
(A) $\dfrac{\sqrt{3}\hat{i}-\sqrt{2}\hat{j}-\sqrt{2}\hat{k}}{\sqrt{8}}$

(B) $\dfrac{\sqrt{3}\hat{i}-\sqrt{3}\hat{j}-\sqrt{2}\hat{k}}{\sqrt{8}}$

(C) $\dfrac{\sqrt{2}\hat{i}-\sqrt{3}\hat{j}-\sqrt{3}\hat{k}}{\sqrt{8}}$

(D) $\dfrac{\sqrt{2}\hat{i}-\sqrt{3}\hat{j}-\sqrt{3}\hat{k}}{\sqrt{8}}$

5. At t = 0, source starts accelerating with an acceleration a and observer starts moving with constant velocity v_o as shown in the figure simultaneously. Source emits a frequency f and velocity of sound in the air is v. The frequency of the initial sound as detected by the observer is.



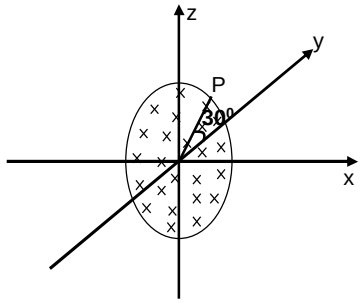
(A) $\dfrac{(v-v_0)f^2}{(2vf+a)}$

(B) $\dfrac{2(v-v_0)f^2}{(2vf+a)}$

(C) $\dfrac{(v-v_0)f^2}{(2vf-a)}$

(D) $\dfrac{2(v-v_0)f^2}{(2vf-a)}$

6. A uniformly charged non-conducting circular disc is placed in the y-z plane with its centre at origin. Its surface charge density is σ C/m^2 and radius R meter. A unit positive charge is brought very slowly from infinity to the point P on its circumference (as shown in figure). The work done (in Joule) by the external agent in this process is.



(A) $\dfrac{\sigma R}{2\varepsilon_0}$

(B) $\dfrac{\sqrt{3}\sigma R}{2\varepsilon_0}$

(C) $\dfrac{\sigma R}{\sqrt{3}\varepsilon_0}$

(D) $\dfrac{\sigma R}{\pi\varepsilon_0}$

(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

7. The two endpoints of an 8 m long weightless thread are fixed at the same height at a distance of 4 m. An object of mass 10 kg is hung onto the thread and can move without friction. A horizontal force of magnitude F is exerted on the object to keep it at rest on the thread exactly below one of the endpoints of the thread? If F is removed, the object has an initial acceleration 'a' and eventually acquires a maximum speed v_{max}.

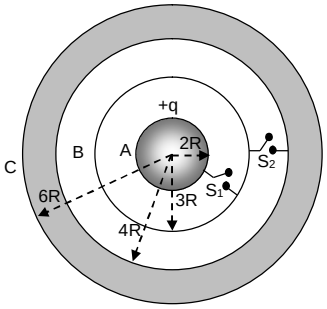
(A) F = 50 N

(B) $a = 4\ m/s^2$

(C) $v_{max} = \sqrt{20\left(2\sqrt{3}-3\right)}$ m/s

(D) The radius of curvature of path traced by the particle after removal of F will first decrease and then increases

8. Three conducting bodies A (solid sphere), B(thin spherical shell) and C(hollow sphere with thickness 2R) are arranged as shown in the figure. A charge q is given to the inner sphere. Choose the correct options(s).



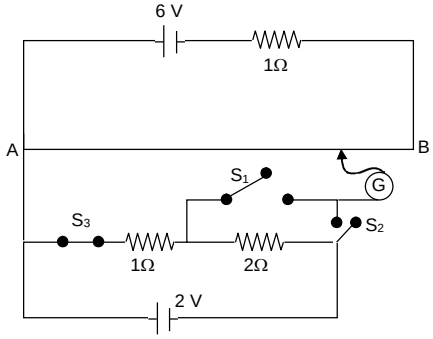
(A) When S_1, S_2 both are open capacity of the system is C_1, then C_1 is $\frac{48\pi\varepsilon_0 R}{5}$

(B) When S_1 is closed but S_2 is open capacity is C_2, then C_2 is $16\pi\varepsilon_0 R$

(C) When S_2 is closed but S_1 is opened capacity is C_3, then C_3 is $12\pi\varepsilon_0 R$

(D) When both S_1 and S_2 are closed. Capacity is C_4, then C_4 is $24\pi\varepsilon_0 R$

9. Figure shows a potentiometer circuit. The length of potentiometer wire AB is 120 cm and its resistance is 2Ω. The internal resistance of 2V cell is 1 Ω initially the switches S_1 and S_2 are open and S_3 is closed.



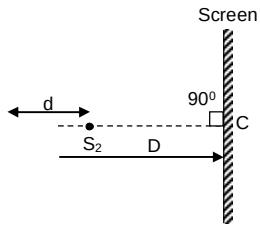
(A) When only S_2 is open, balanced length $\ell = 20$ cm

(B) When only S_1 is open, balanced length $\ell = 60$ cm

(C) When all three switches are closed, balanced length is 60 cm

(D) When S_3 is open, S_2 is also open but S_1 is closed, balance length is 60 cm

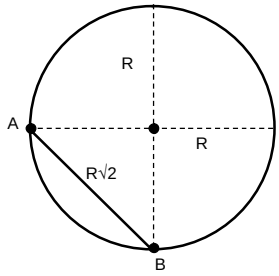
10. Two coherent sources are emitting light of wavelength λ = 5000 Å. The separation between the sources is 2.5 mm and the separation between the source and the screen is 2m. then which of the following are correct.



(A) The position of 3000th order bright fringe from C is 8/3 m.
(B) The position of 2500th order bright fringe from C is $2\sqrt{3}$ m.
(C) The position of 5th bright fringe from C is 0.08 m (include the bright fringe at C).
(D) The shape of fringes on the screen are circular.

11. One mole of an ideal monoatomic gas undergoes a process $T = \left(\frac{T_0}{V_0^2}\right)V^2$, where T_0 and V_0 are the initial temperature and volume of the gas. If volume of gas changes from V_0 to $2V_0$. The choose the correct option(s).
(A) The amount of heat transferred to the gas is $6RT_o$.
(B) The molar heat capacity of the gas is 2R.
(C) Final temperature of the gas is $4T_o$.
(D) The graph between the temperature and volume is parabolic.

12. Two small balls 'A' and 'B' of masses 0.3 kg and 0.6 kg respectively are rigidly attached to the ends of a light rigid rod of length $R\sqrt{2}$ and placed inside a fixed smooth spherical cavity of radius R = 0.5 m with the ball 'B' is located at the lowest point of cavity as shown in the figure. Then choose the correct option(s) immediately after the rod is released from rest. (Take g = 10 m/s^2)

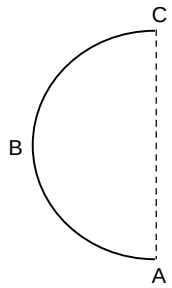


(A) The normal force on the ball 'A' due to surface of cavity is 2N.
(B) The normal force on the ball 'A' due to surface of cavity is 4N.
(C) The normal force on the ball 'B' due to surface of cavity is 8N.
(D) The force acting on the ball 'B' due to light rod is $2\sqrt{2}$ N.

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

13. A semicircular conducting ring ABC of mass m = 20 kg and radius r = 4m is placed on a horizontal surface and a uniform magnetic field B = 2 Tesla exists parallel to horizontal surface but perpendicular to the line AC. Almost instantaneously, a certain amount of charge is passed through it, then it is found to jump to a height h = 5m. then, find the amount of charged passed through the conductor is coulomb (take g = 10 m/s^2)

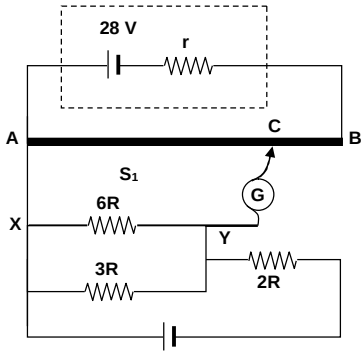


14. A certain radio nuclide starts getting produced at a constant rate 10^4 sec^{-1} by some mechanism at a certain moment t = 0. At this moment there are already 10^5 number of radio nuclides present in the sample. The mean life of this radio-nuclide is 20 seconds. At t = 40 sec, the $\frac{dN}{dt}$ is $\frac{10^4}{K}$ (in sec^{-1}). then find the value of K. (Take $e^2 = 7.4$)

15. Peak emission from a black body at a certain temperature occurs at a wavelength of 4000Å. On increasing its temperature, the total radiation emitted is increased to 256 times. These radiations are allowed to fall on a metal surface. The photoelectrons emitted by peak emission wavelength at higher temperature can be brought to rest by applying a potential which is equivalent to the excitation potential corresponding to the transition from level n = 2 to level n = 4 in the singly ionized Helium atom (obeys Bohr's Model). The work function of the metal (in eV) is.

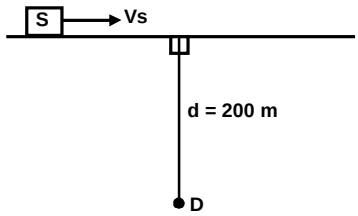
[Given E (in eV) = $\frac{12400}{\lambda\left(\text{in } \overset{0}{\text{A}}\right)}$]

16. In the given potentiometer circuit, the resistance per unit length (in Ω/m) 'λ' of the potentiometer wire varies with distance x (from end A) as λ = 1 + 2x. The length of the potentiometer wire is 1 m. While measuring the potential difference between the points X and Y, the balancing length (AC) is obtained exactly at the middle of the wire. The internal resistance (r) of the driving cell (in ohm) is



17. At what speed (in m/s) must a pebble be thrown from a height of h = 3.6m, and at an angle of 30^0, measured from the horizontal, if it is to hit the ground at an angle of 45^0. (Air drag can be neglected.) (Take g = 10 m/s^2)

18. A source emitting sound of frequency 600 Hz is moving along a straight horizontal road with a constant velocity 66 m/s. A stationary detector 'D' is located at a distance 200 m from the straight road. The velocity of sound in air is 330 m/s. Find the frequency (in Hz) of sound received by the detector when the source is closest to the detector.



Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **06 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

19. Among the following statements which is incorrect?
    (A) In the preparation of compounds of Xe, Bartlett had taken O_2PtF_6 as a base compound because both O_2 and Xe have almost same ionization enthalpy.
    (B) Nitrogen does not show allotropy.
    (C) A brown ring is formed in the ring test for NO_3^- ion. It is due to the formation of $\left[Fe(H_2O)_5NO\right]^{\oplus 2}$
    (D) On heating with concentrated NaOH solution in an inert atmosphere of CO_2, red phosphorus gives PH_3 gas.

20. For a certain reaction, the variation of the rate constant with temperature is given by the equation $\ln K_t = \ln K_0 + \frac{t}{10}(\ln 2),\ (t \geq 0^\circ C)$. The value of temperature coefficient (T_C) of the reaction rate is therefore
    (A) 4
    (B) 2
    (C) 3
    (D) 1

21. Which of the following statement is incorrect?
    (A) Be dissolves in alkali forming $\left[Be(OH)_4\right]^{-2}$
    (B) Isoelectric point (PI) of amino acids calculated by using relation, $PI = \frac{pK_{a_1} + pK_{a_2}}{2}$ only.
    (C) Alkaline earth metal ion, because of their much larger charge to size ratio exert a much stronger electrostatic attraction on the oxygen of water molecule surrounding them
    (D) BeF_2 form complex ion with NaF in which Be goes with anion.

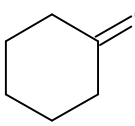
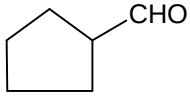
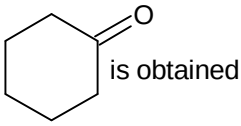
22. In diazotization of aniline with sodium nitrite and hydrochloric acid, an excess of Hydrochloric acid used primarily to
    (A) Suppress the concentration of free aniline available for coupling
    (B) Suppress hydrolysis of phenol.
    (C) Ensure a stoichiometric amount of nitrous acid
    (D) Neutralize the base liberated

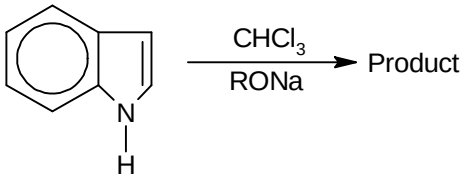
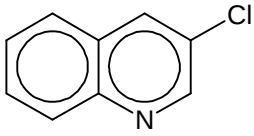
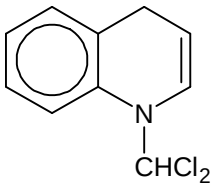
23. In an experiment based on graham's law of diffusion. It is observed that the speed ratio of diffusion of gases x and y is 2:4. If the ratio of their masses present in the mixture is 2:3, then the ratio of their mole fractions is
    (A) 1:6
    (B) 3:2
    (C) 2:3
    (D) 1:24

24. The room energy per mole on a cold winter day at $-8^{o}C$ nearly
(A) 3.9 kJ/mol
(B) 2 kJ/mol
(C) 3.3 kJ/mol
(D) 10 kJ/mol

(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

25. Cis-2-amino cyclohexanol $\xrightarrow{HNO_2} A + B,$ which statement(s) is/are correct regarding reaction?
(A) proceed via diazotization
(B) N_2 gas evolved in Intermediate step
(C) Products are  and 
(D) only  is obtained

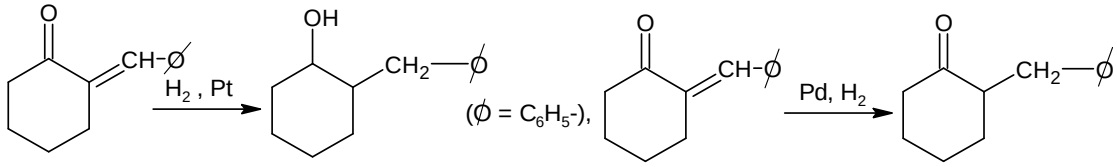
26. 
Select the correct option(s) based upon the above given reaction:
(A) The intermediate involved in the reaction is dichloromethylene
(B) The magnetic moment (spin only) for intermediate of the reaction acting as weak electrophile may be $\sqrt{8}$ B.M.
(C) Major prodcut is 
(C) Major prodcut will be 

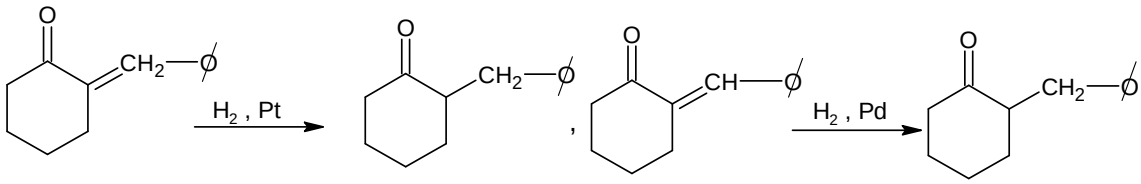
27. A compound (x) of S, Cl and O has vapour density of 67.5, it reacts with water to form two acids, and reacts with KOH to form two salts (y) and (z). While (y) gives white precipitate with $AgNO_3$. (z) gives white precipitate with $BaCl_2$ solution. Select the correct statement(s) is/are
(A) x is $SOCl_2$ which on hydrolysis gives HCl and SO_3.
(B) x is SO_2Cl_2 which on hydrolysis gives HCl and H_2SO_4
(C) y is KCl which with $AgNO_3$ gives AgCl
(D) z is H_2SO_4 which with $BaCl_2$ gives $BaSO_4$.

28. Select the correct statement(s):

(A) A compound having enantiomers must be chiral

(B) A compound having enantiomers must have chiral centres

(C) 

(D) 

29. Which of the following relations are correct for an ideal gas?

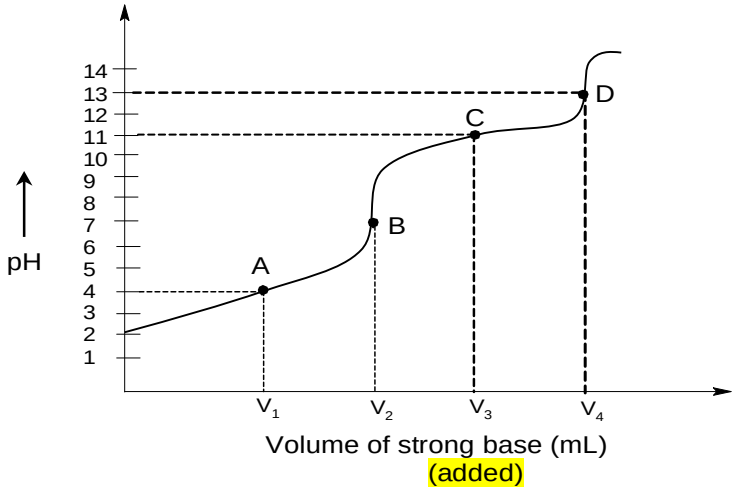
(A) $\left(\frac{dE}{dT}\right)_T = 0$

(B) $\left(\frac{dH}{dT}\right)_T = 0$

(C) $\left(\frac{dG}{dT}\right)_P = -S$

(D) $\left(\frac{dG}{dP}\right)_T = V$

30. A weak diprotic acid $H_2A\,(.1\,M,\ 25\ mL)$ is titrated against strong base (.1 M) as



The correct statement(s) is/are

(A) pK_{a_1} and pK_{a_2} of H_2A is 4 and 11 respectively when $V_1 = \frac{V_2}{2}$ and $V_3 = \frac{3V_2}{2}$.

(B) at point B an amphiprotic anionic species is present in the solution

(C) at point C, pH of solution will be approximately $= pK_{a_2} + \frac{\left[A^{-2}\right]}{\left[HA^{-}\right]}$.

(D) at point D, pH of solution calculated by salt hydrolysis phenomena.

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

31. The reaction of NH_3 with dimer of BH_3 gives initially compound 'x' further on heating 'x' gives a compound 'y' which behaves like an aromatic compound. The no. of Ammonia molecules present in cationic part of compound 'x' is 'e'. Calculate the value of 'a', $a = \frac{e + f + 15}{2}$ where f is steric factor of anionic part.

32. The ionization energy of lithium is 5.40 eV/atom. If ionization energy of H is 13.6 eV/atom, then effective nuclear charge acting upon outer most electron of Li is

33. The Boiling point of an aqueous solution is 100.1°C calculate its freezing point. Given that latent heat of fusion and latent heat of vaporization of water is 80 calorie/gram and 540 calorie/gram respectively.

34. At 291 K, the molar conductivities at infinite dilution of NH_4Cl, NaOH and NaCl are 129.8, 217.4 and 108.9 $scm^2\ mol^{-1}$ respectively. If molar conductivity of a centinormal solution of NH_4OH is 9.33 $scm^2\ mol^{-1}$, What is the percentage dissociation of NH_4OH at this dilution?

35. The rate law for the reaction of $\overline{O}H$ with tert-butyl bromide to form an elimination product in 75% ethanol / 25% water at 30° C is the sum of the rate laws for the E_2 and E_1 reactions:
Rate = 7.1×10^{-5} [tert-butyl bromide] $\left[\overline{O}H\right] + 1.5\times10^{-5}$ [tert-butyl bromide]. If the percentage of the reaction takes place by the E_2 path in $\left[\overline{O}H\right] = 5$ M is x, then value of $\frac{x}{3.5}$ is

36. A metal crystallizes into two phases fcc and Bcc whose unit lengths are 3.5 A° and 3 A° respectively. If the ratio of densities of fcc to bcc is x. Then calculate the value of a, a = x + y (y is the number of unpaired electron in $\left[NiF_6\right]^{-2}$ ion)

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **06 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. Suppose that f is differentiable for all x and that $f'(x) \le 2$ for all x. If f(1) = 2 and f(4) = 8. Then f(2) has the value equal to
(A) 3
(B) 4
(C) 6
(D) 8

38. Number of integral values of a for which every solution of the inequality $x^2 - 3x + 4 > 0$ is also the solution of the inequality $(a-1)x^2 - (a + |a-1| + 2)x + 1 \ge 0$ is
(A) 0
(B) 1
(C) 2
(D) 3

39. If a, b, c are distinct positive real numbers in H.P., then equation $x^2 - kx + 2b^{101} - a^{101} - c^{101} = 0$ $(\text{where } k \in R)$ has
(A) Both roots positive
(B) one positive and one negative root
(C) Both roots negative
(D) Both roots imaginary

40. Let $f(x) = \sin^{-1} x - x$ and $g(x) = x^2 + 5x + 6 + \cos^{-1} x$. If $|f(x)| + |g(x)| = |f(x) + g(x)|$, then true set of real values of x, is
(A) $\{x : x \in R, -1 \le x \le 1\}$
(B) $\{x : x \in R, -1 \le x \le 0\}$
(C) $\{x : x \in R, 0 \le x \le 1\}$
(D) $\left\{x : x \in R, -\frac{1}{2} \le x \le \frac{1}{2}\right\}$

41. Let $f : R \to R$ be a continuous onto function satisfying $f(x) + f(-x) = 0 \forall x \in R$. If $f(-3) = 2$ and $f(5) = 4$ in $[-5,5]$, then the equation f(x) = 0 has:
(A) exactly three real roots
(B) exactly two real roots
(C) atleast five real roots
(D) atleast three real roots.

42. Let $n \in I^+$ and consider the point P with abscissa 'n' on the curve $y^2 - x^2 = 1$. If d_n represents the shortest distance from the point P to the line y = x then $\lim\limits_{n\to\infty}(n.d_n)$ has value equal to

(A) $\frac{1}{2}$

(B) $\frac{1}{\sqrt{2}}$

(C) $\frac{1}{2\sqrt{2}}$

(D) 0

(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

43. If $f''(x) > 0$ for all $x \in [-3,4]$, then which of the following are always true?

(A) f(x) has a relative minimum on (-3, 4)

(B) f(x) has a minimum on [-3, 4]

(C) f(x) is concave upwards on [-3, 4]

(D) If $f(3) = f(4)$ then $f(x)$ has a critical point on $[-3,4]$

44. A line L passing through the point P(1, 4, 3) is perpendicular to both the lines $\frac{x-1}{2} = \frac{y+3}{1} = \frac{z-2}{4}$ and $\frac{x+2}{3} = \frac{y-4}{2} = \frac{z+1}{-2}$. If the position vector of point Q on L is (a_1, a_2, a_3) such that $(PQ)^2 = 357$, then $(a_1 + a_2 + a_3)$ can be

(A) 1

(B) 2

(C) 15

(D) 16

45. Let $f : A \to B$ be a function where set A contains 4 elements and set B contains 3 elements. Number of functions defined from $A \to B$ which are not surjective is also equal to

(A) Number of natural solution of the equation $x + y + z = 11$.

(B) Number of ways in which 10 children can be divided into two groups one containing 2 and other containing 8 children.

(C) Number of ways in which 4 boys and 4 girls can be arranged alternately in a circle.

(D) Total Number of divisors of the number 3600.

46. If $f(x) = \text{sgn}\left(\left(\sin^2 x - \sin x - 1\right).\left(\sin^2 x + \sin x + 1\right)\right) = 0$ has exactly four points of discontinuity for $x \in (0, n\pi), n \in N$, then the value of n may be equal to

(A) 2

(B) 3

(C) 4

(D) 5

47. A function y = f(x) satisfying the differential equation $\frac{dy}{dx}.\sin x - y\cos x + \frac{\sin^2 x}{x^2} = 0$ is such that, $y \to 0$ as $x \to \infty$, then the statement which is correct, is/are

(A) $\lim\limits_{x\to 0} f(x) = 1$

(B) $\int_0^{\pi/2} f(x)dx$ is less than $\pi/2$

(C) $\int_0^{\pi/2} f(x)\,dx$ is greater than unity

(D) f(x) is an odd function

48. Let $f:(0,\infty)\to(-\infty,\infty)$ be defined as $f(x)=e^x+\ln x$ and $g(x)=f^{-1}(x)$, then

(A) $g''(e)=\frac{1-e}{(1+e)^3}$

(B) $g''(e)=\frac{e-1}{(e+1)^3}$

(C) $g'(e)=e+1$

(D) $g'(e)=\frac{1}{e+1}$

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. Let f(x) be a polynomial of degree 6 divisible by x^3 and having a point of extremum at x = 2. If f'(x) is divisible by $(1+x^2)$, then the value of $\frac{f(2)}{f(1)}$ is

50. If $f^2(x)+g^2(x)+h^2(x)\le 9$, where f(x), g(x) and h(x) are continuous for all $x\in R$ then find maximum value of $(3f(x)+4g(x)+10h(x))$.

51. Let $f:R\to R$ be an injective mapping and p, q, r are non-zero distinct real quantities satisfying $f\left(\frac{p}{r}\right)=f\left(\frac{p-q}{q-r}\right)$ and $f\left(\frac{q}{r}\right)=f\left(\frac{r}{p}\right)$. If graph of $g(x)=px^2+qx+r$ passes through point (1, 6), then the value of q is

52. Let A be 3×3 matrix given by $A=[a_{ij}]$ and B be a column vector such that B^TAB is a null matrix for every column vector B. If $C=A-A^T$ and $a_{13}=1$, $a_{23}=-5$, $a_{21}=15$, then the value of det (adj A) + det (adj C) is

53. If $f:R^+\to R$ be a differentiable function satisfying $f(x)=e+(1-x)\ln\left(\frac{x}{e}\right)+\int_1^x f(t)dt\ \forall x\in R^+$, then the area enclosed by the curve $g(x)=x(f(x)-e^x)$ lying in the fourth quadrant is equal to

54. If $|g(x)-g(y)|\le|x-y|\ \forall\ x,y\in[2,10]$, then the maximum value of $\left|\int_2^{10} g(x)dx-8g(\alpha)\right|$ for $\alpha\in[2,10]$ is equal to